Nowcast Lab

Crude oil, alternative data and macro nowcasts from free public data

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30 October 2026 · data as of 2026-10-27

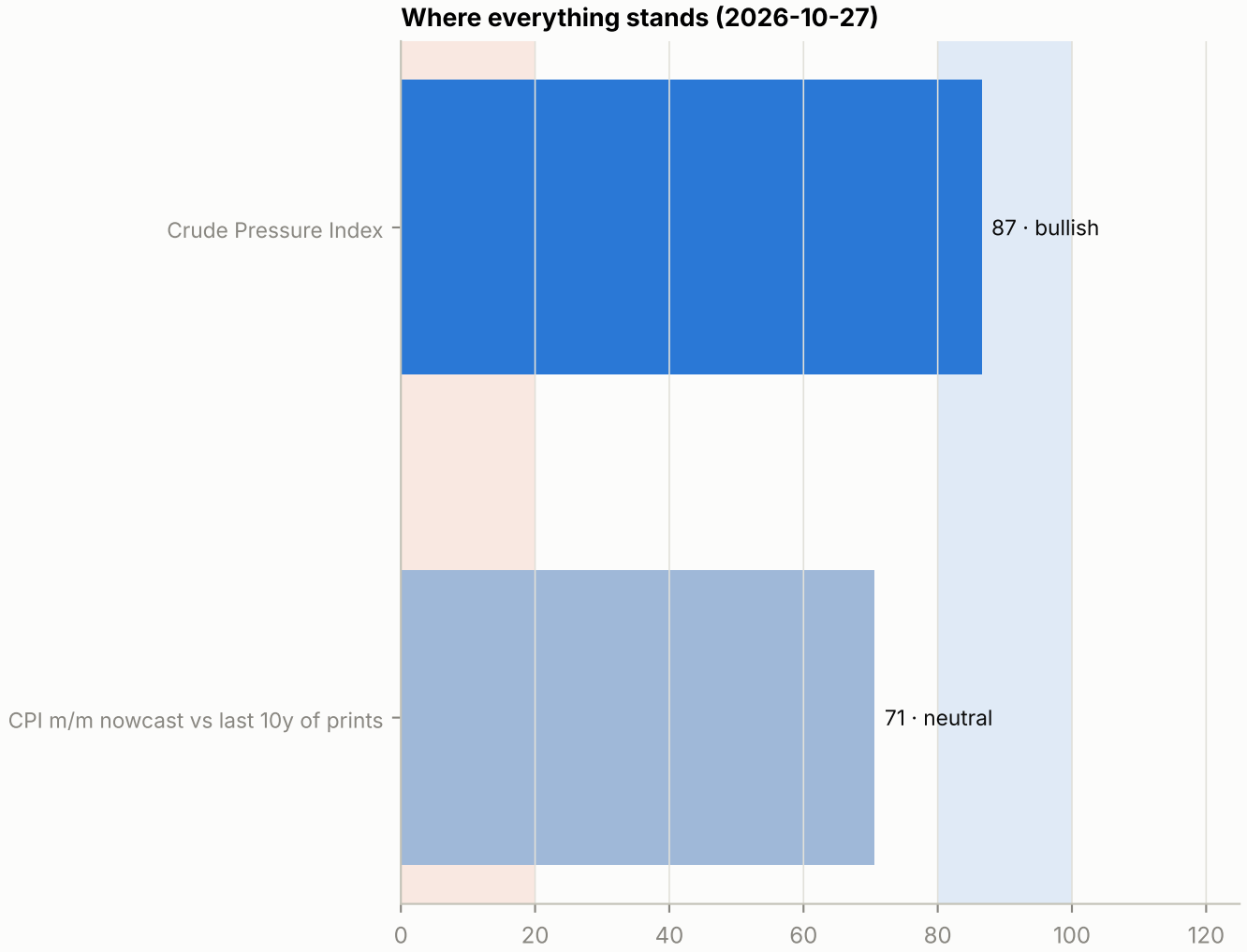
Crude and dollar data: Jan 1986 to Oct 2026 (9,532 obs)

US inflation: Jan 1913 to Aug 2026 (1,363 obs)

Transparent models, publication-aware timing, baselines first, walk-forward tests. Context signals, not trading advice.

Summary: where things stand today

Readings as of 2026-10-27, each against its own trailing three years



Reading today.

The macro backdrop is inside its normal range on every family shown.

- Crude one-week view: +0.67% expected, out-of-sample R^2 0.69%, Diebold-Mariano t 0.5: no proven edge over a random walk.
- CPI nowcast (2026-10): +0.43, error -17% vs baseline, t 2.9 (significant).

Data sources (1/2): crude oil and alternative data

Free public sources; period shown is what was actually loaded in this run

Series	Source	Frequency	Usable from	Period loaded
WTI crude spot price	US EIA (RWTC), via GitHub datasets/oil-prices	daily	same day	Jan 1986 to Oct 2026 (9,532 obs)
Brent crude spot price	US EIA (RBRTE), via GitHub datasets/oil-prices	daily	same day	May 1987 to Oct 2026 (9,117 obs)
US dollar vs 8 currencies	Federal Reserve H.10, via GitHub...	daily (released...	usable from Tuesday close	Jan 1995 to Oct 2026 (7,981 obs)

Alternative data (tankers, EIA weekly, positioning, JODI, news) was not loaded in this run, so those slides are left out.

Data sources (2/2): macro and market series

US data from FRED and the BLS; period shown is what was actually loaded in this run

Series	Source	Frequency	Usable from	Period loaded
US consumer prices (CPI)	US BLS, via GitHub datasets/cpi-us	monthly	month end +17 days	Jan 1913 to Aug 2026 (1,363 obs)

Test periods: crude index scored out-of-sample from 2008; monthly macro nowcasts walk forward from 2008 (96-month minimum training); standardisation uses trailing 3 years; correlation map uses the last 3 years.

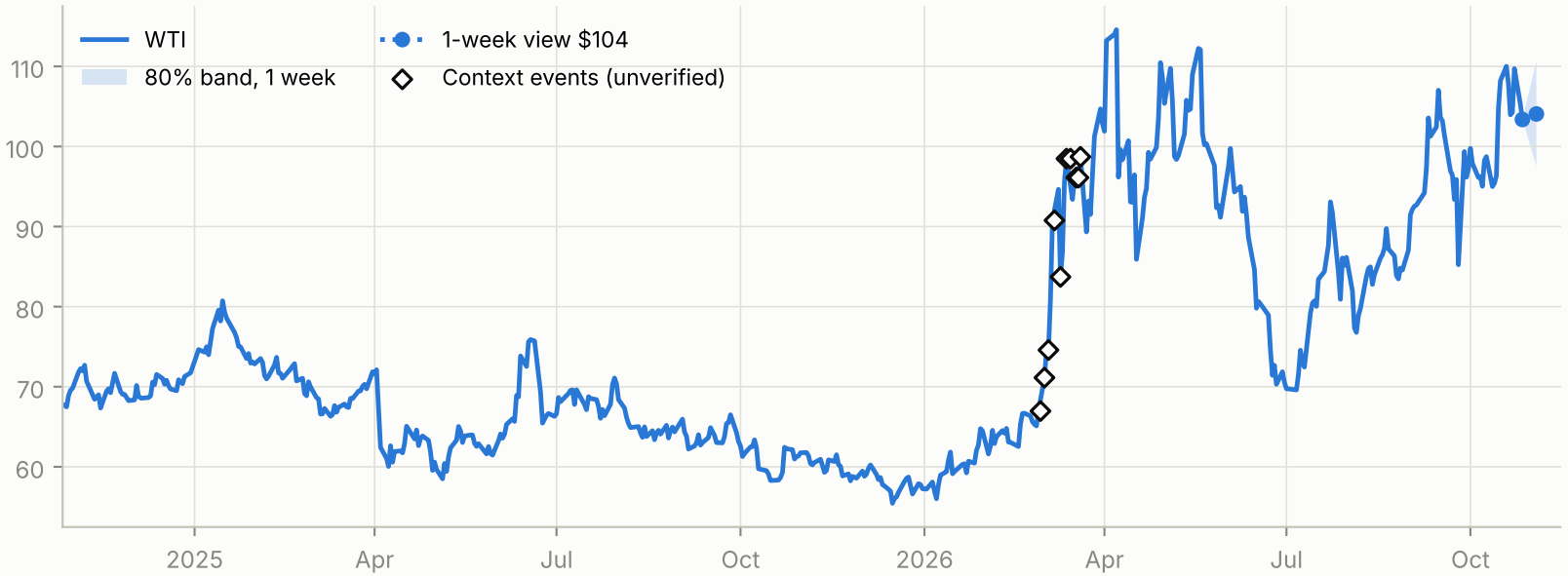
Method and look-ahead checks

How the numbers are built

- Publication-aware: every series enters only from the day it was actually released (CPI about 17 days after month end, jobs about 8 days, claims 5 to 12 days, daily market data next day). Nothing is used before it was public.
- Trailing windows only: each signal is standardised against its own previous three years (min one year), clipped at ±3 standard deviations. The index is a fixed-weight sum, with weights set before testing.
- Baselines first: a model only counts if it beats a simple baseline (random walk for prices; seasonal, last-published and 12-month average for monthly data).
- Walk-forward testing: models are refit on past data only, then scored on the next period. Significance uses the Diebold-Mariano test with Newey-West errors; |t| below 2 means no proven edge. Covid months (2020-03 to 2021-06) are reported separately.
- Multiple testing: the alternative-data signals are tested many times at once, so p-values are Bonferroni-adjusted before any signal is trusted.
- Look-ahead checks: rebuilding the features as of 4 past dates gave identical values at every date, so no future data leaked; a planted-leak test confirms the checker would catch one.

Crude Pressure Index and one-week view

WTI spot, USD per barrel (last 2 years)



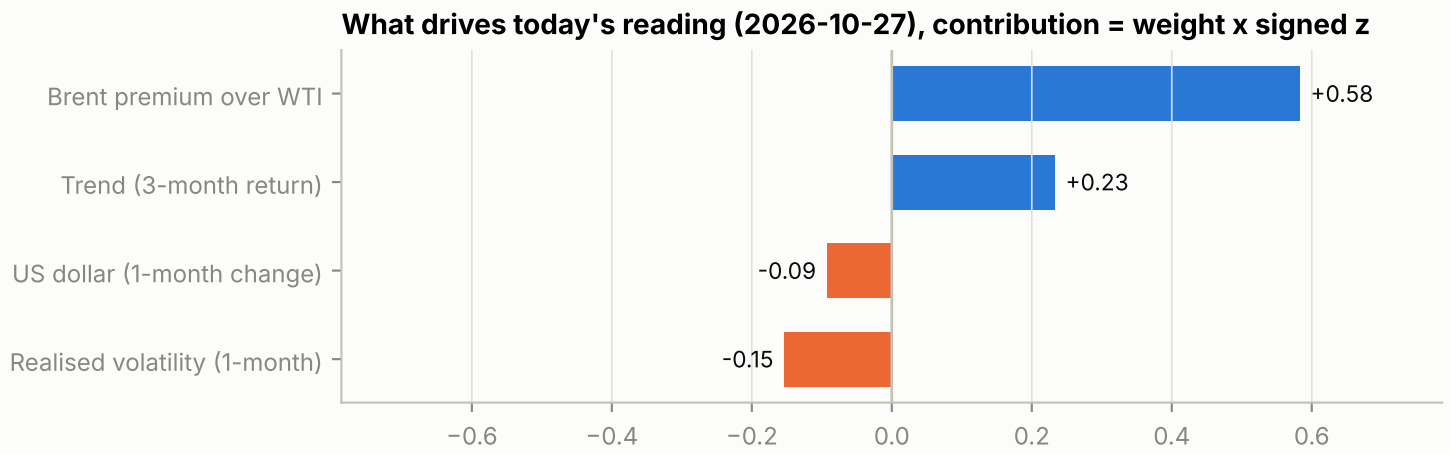
Crude Pressure Index, percentile vs trailing 3 years (shaded: top and bottom 20%)

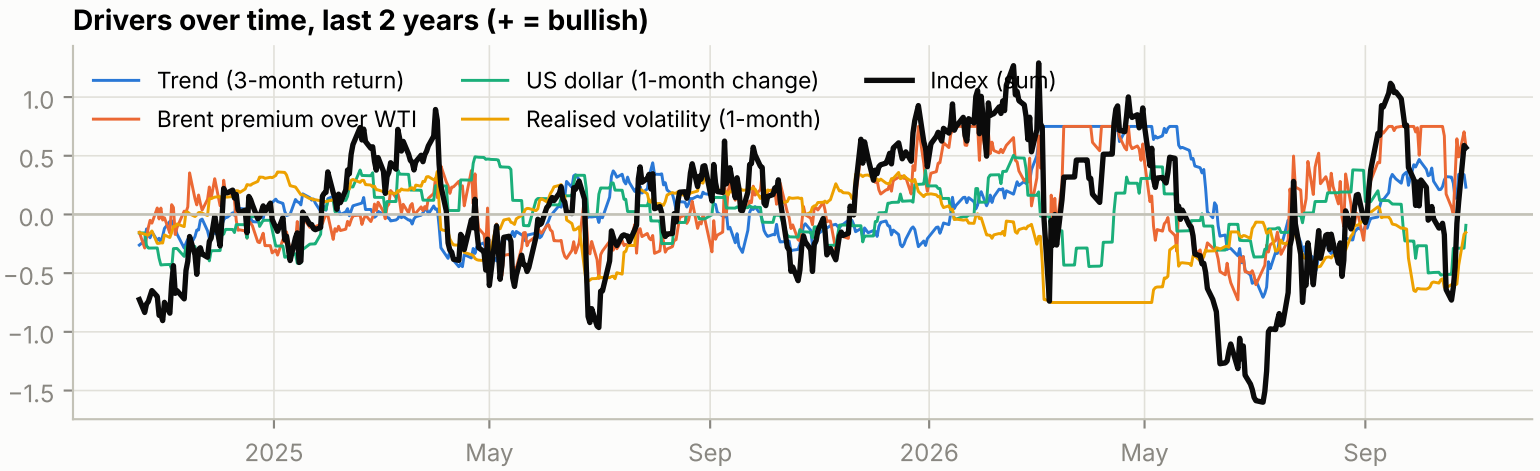


Reading the history chart.

Over the last five years the index was in the bullish zone on 25% of days and in the bearish zone on 20%. Its highest reading was on 2026-03-03 and its lowest on 2026-07-06. Over the last 21 trading days it moved +0.49, so pressure is currently building. A level only means something relative to its own trailing history, which is why the zones use percentiles, not fixed thresholds. The index describes pressure on prices; it is not a price forecast on its own.

What drives the index



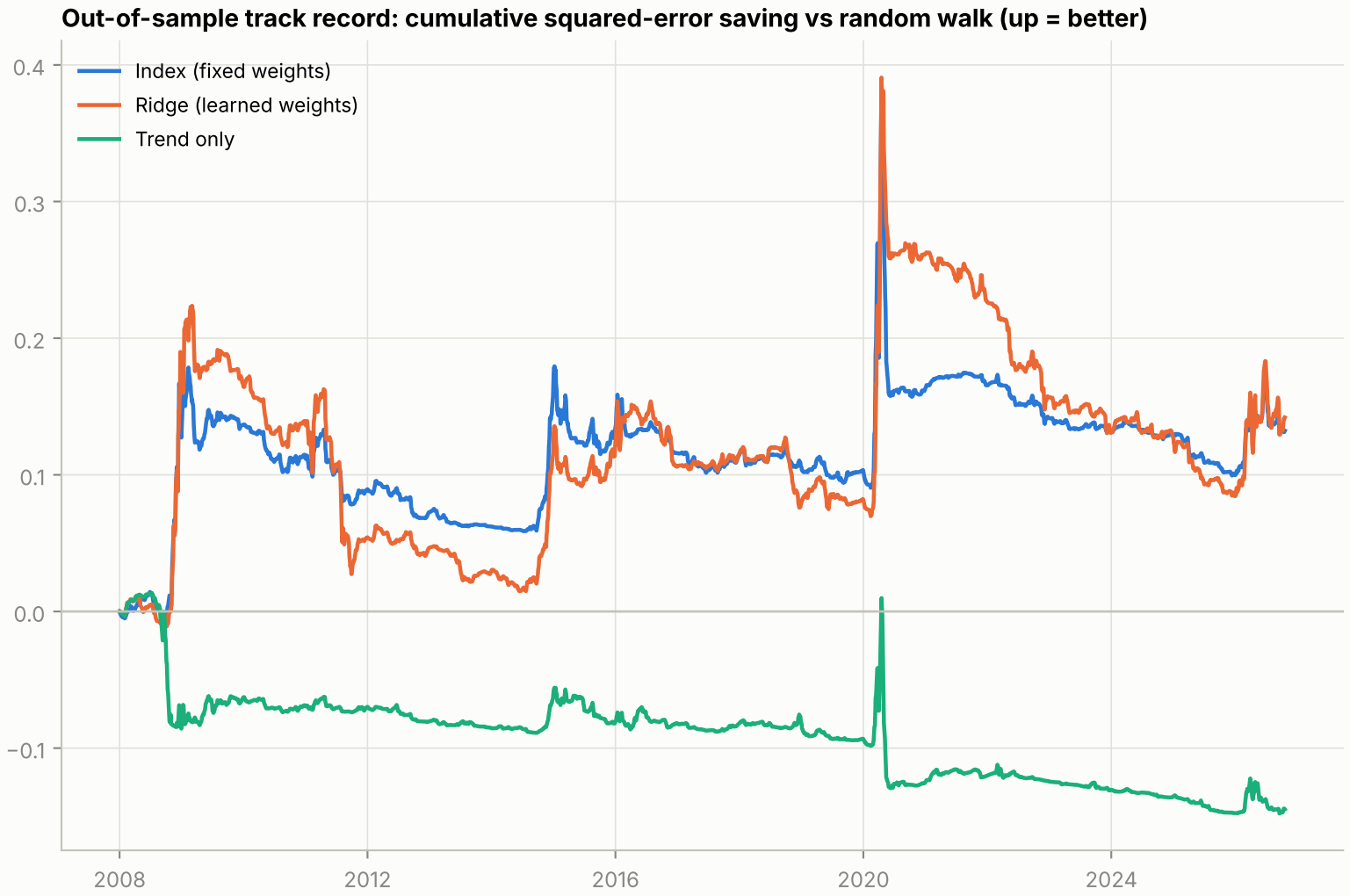


Reading the drivers chart.

Each bar is one component's contribution to today's index (+0.57); green pushes toward higher prices, red toward lower. The largest pushes are Brent premium over WTI (+2.3 σ, bullish) and Trend (3-month return) (+0.9 σ, bullish). The index sits at the 86th percentile of its trailing three years, so it sits in the top band (≥ 80), the bullish zone.

The one-week view this implies is +0.67% (WTI 103.37 → about 104.06, 80% range 97.56–110.67). Context signal, not a trading recommendation. The one-week view is not statistically distinguishable from a random walk (|DM t| < 2).

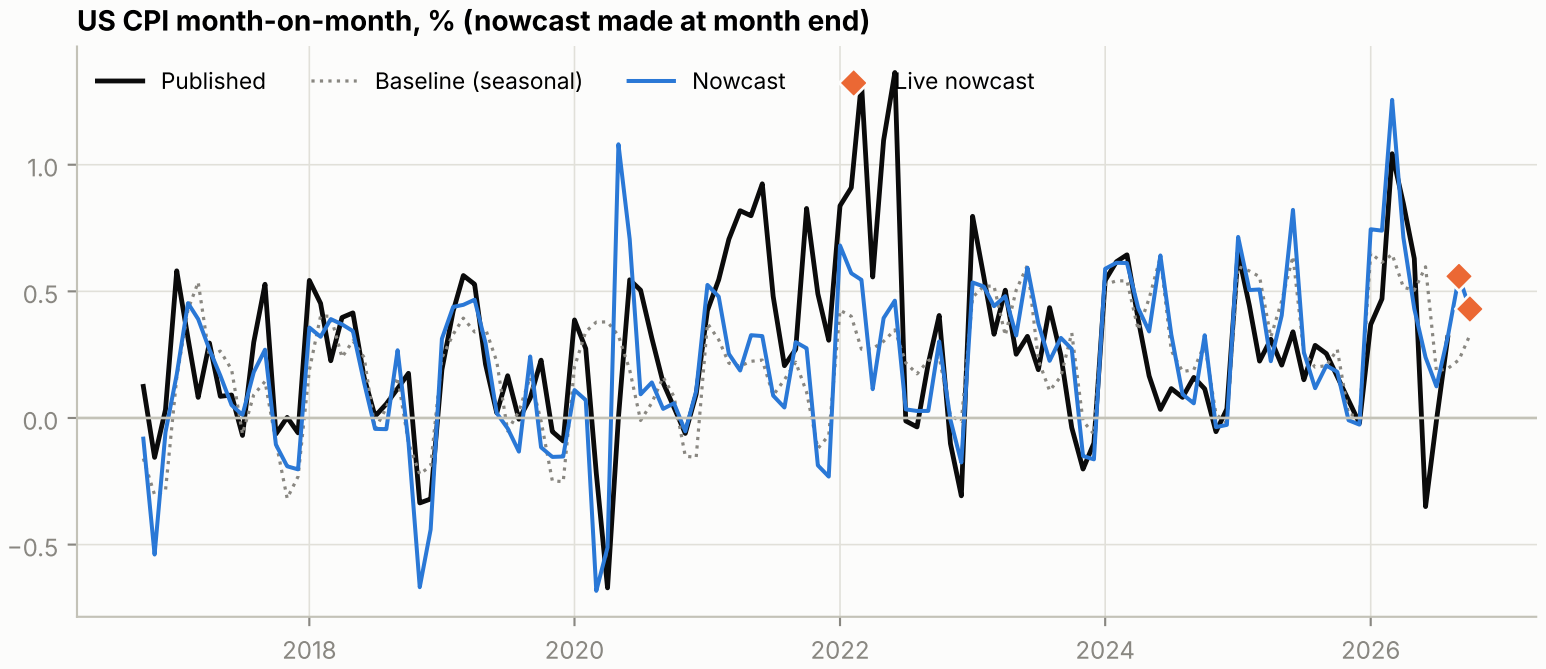
Does it beat a random walk? Out-of-sample track record

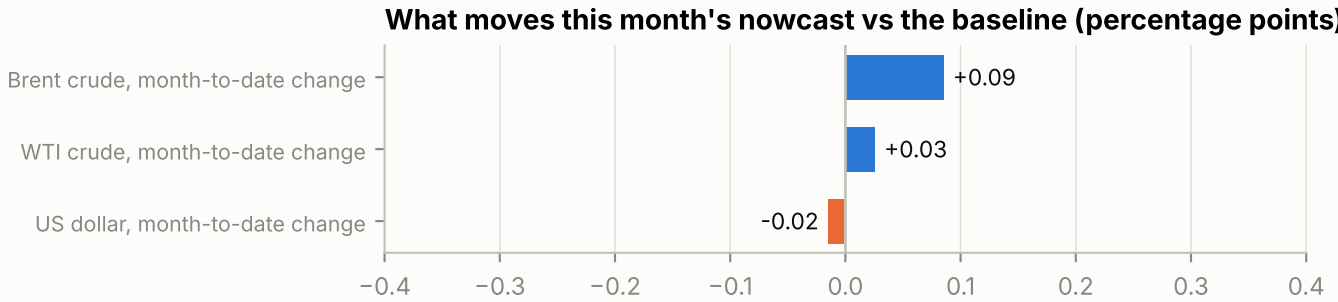


Reading the skill chart.

The line adds up the squared-error saving of the index model over a random walk; a rising line means the model beat "no change", a falling line means it lost. Over the full out-of-sample period the one-week out-of-sample R^2 is 0.69% with a hit rate of 52.2% and rank correlation 0.027; against the random walk it is not distinguishable from the baseline (t = 0.5, inside ±2). It beat the random walk in 6 of 19 calendar years (best 2014: +29.0%, worst 2012: -6.4%). A line that gains in a few bursts and is flat elsewhere means the result depends on a few episodes, not on steady skill.

Inflation: US CPI month-on-month nowcast





Reading the CPI chart.

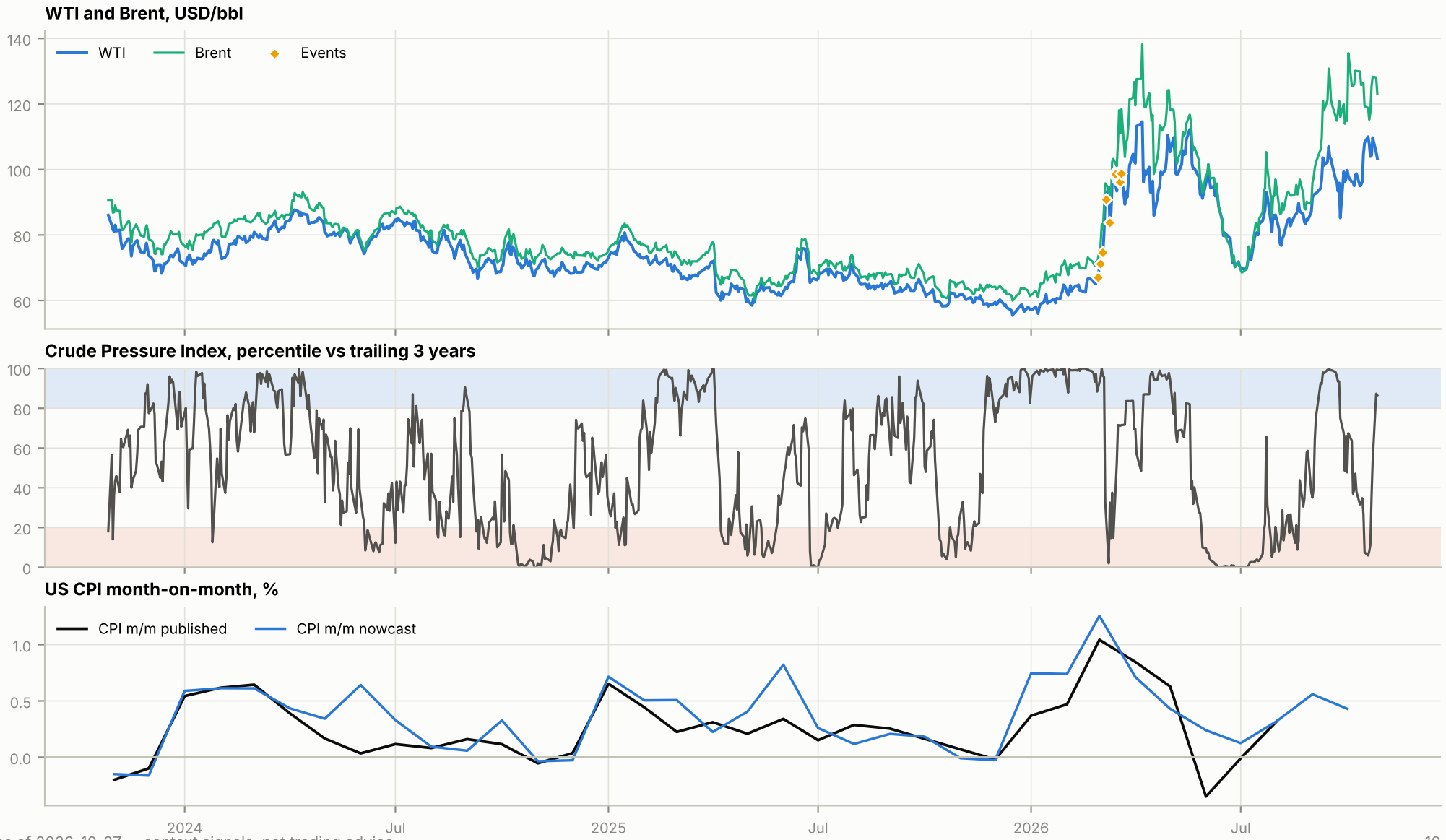
Black is what was published, the dotted line is the baseline (seasonal), blue is the nowcast made using only data available at the time. Excluding Covid, the nowcast's error is -17% versus the baseline over 206 months (negative = smaller error), and statistically it is significantly better (t = 2.9). Made at mid-month, with less information, the figure is -17% (t = 3.2). The largest misses in the plotted window were May 2020 [Covid window] (actual +0.00%, nowcast +1.08%) and Jun 2022 (actual +1.36%, nowcast +0.46%). During the 2021–23 inflation surge the nowcast was on average 0.27 points too low per month: the model only captures how energy prices pass through to CPI, so inflation driven by rents and core goods is outside what it can see.

Today: the nowcast for 2026-10 is +0.43% against a baseline of +0.33%, a gap of +0.10. That gap is small relative to the typical miss, so treat it as a tilt, not a call.

Reading the drivers chart. Bars show how much each input moves this month's CPI nowcast away from its starting point; green bars push it up, red bars pull it down. The biggest single input is Brent crude, month-to-date change (+0.09), then WTI crude, month-to-date change (+0.03). Together the inputs add up to +0.10.

Everything on one timeline

Crude, alternative data and the macro backdrop share one date axis (last 3 years)



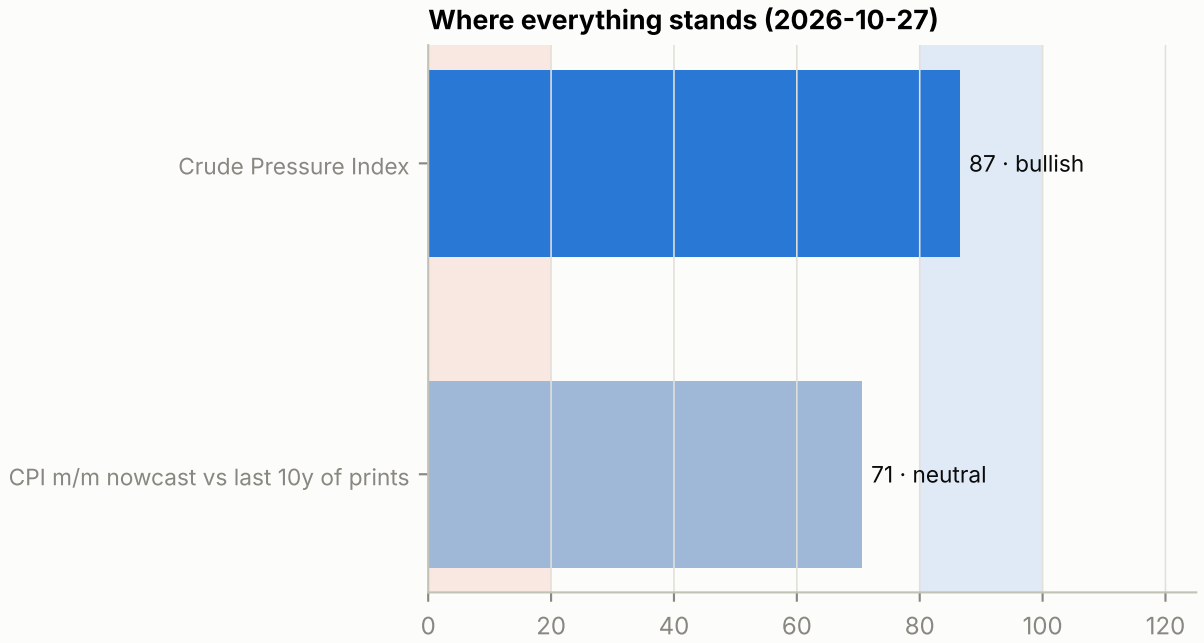
Reading the combined view

How to read the timeline.

All panels share one date axis, so you can look straight down from a price move to see what else was happening. Top to bottom: price (with context events), the Crude Pressure Index, the alternative-data signals (each row one signal, blue = bullish for crude, orange = bearish), their combined average against the index, then the macro backdrop (market stress, labour, inflation). Today the crude index is at the 87th percentile (bullish). No alternative data is loaded in this run, so those panels are omitted.

Backdrop: the CPI nowcast (+0.43% for 2026-10) is neutral versus the last ten years (71th percentile). These sit beside the crude view as context; they are not inputs to it.

Scoreboard and how the pieces move together



Reading the scoreboard. Each bar is the family's current reading as a percentile of its own last three years (the shaded edges are the top and bottom 20% bands). The macro backdrop is inside its normal range on every family shown.

Limits and how to use this

Limits to keep in mind

- Context signals for analysts, not recommendations to trade. The one-week crude view is not statistically distinguishable from a random walk.
- Payroll and unemployment figures are the latest revised values, which flatters those tests slightly.
- The inflation model captures energy pass-through only, so it misses inflation driven by rents and core prices (the 2021 to 2023 surge).
- Alternative-data signals have not been adopted into the index: with about 11 signals at 2 horizons, some will look significant by chance, so only adjusted p-values count.
- The market stress index uses the Baa spread (not high-yield) and is checked against NFCI; S&P 500 history on FRED covers about 10 years.
- News flow (GDELT) is live-only context with no backtest. Context events are a supplied timeline and are not verified.
- Several source endpoints (PortWatch, CFTC, JODI) depend on what was reachable when the notebook ran; the data-source slides show what was actually loaded.

How to read the charts

Reading conventions

- Hover any line for the exact value on that date; drag to zoom, double-click to reset, use the range buttons for 1m / 6m / 1y / all.
- Index percentile (0–100) compares today with the trailing three years only. Above 80 is the top band, below 20 the bottom band; it is a ranking, not a forecast.
- Diamonds mark live nowcasts for a month that has not been published yet. Nothing after the last actual is scored.
- Baselines come first. A nowcast earns attention only if its Diebold–Mariano t is above 2 against the baseline; below that, treat it as context.
- Data lag. Every series appears only from its publication date, so the latest points on weekly and monthly charts are older than the daily ones.
- Everything is context for analysts, not a recommendation to trade.